Pneumonia Detection via Chest X-Ray

Deep Learning Classification Report

Project1.Report 3. Transfer Learning

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1. Introduction

This report presents the design, training, and evaluation of a deep learning model for binary classification of chest X-ray images into two categories: NORMAL and PNEUMONIA. The project compares a custom CNN baseline (Variant B) against a ResNet50 transfer learning approach, with the transfer learning model achieving superior performance across all metrics.

The primary objective is to develop a reliable, clinically applicable classifier that minimises false negatives (missed pneumonia cases) while maintaining high overall accuracy.

2. Methodology

2.1 Model Architecture

The final model uses ResNet50 as a pre-trained backbone, fine-tuned on chest X-ray data in two distinct training phases:

- Phase 1 — Feature Extraction: The ResNet50 base layers were frozen. Only the custom classification head was trained for 20 epochs, allowing the model to adapt the pre-learned ImageNet features to the medical imaging domain.
- Phase 2 — Fine-tuning: The top layers of ResNet50 were unfrozen and retrained with a low learning rate (1e-5), allowing gradual adaptation of deep features to the specific characteristics of chest X-ray images.

2.2 Training Configuration

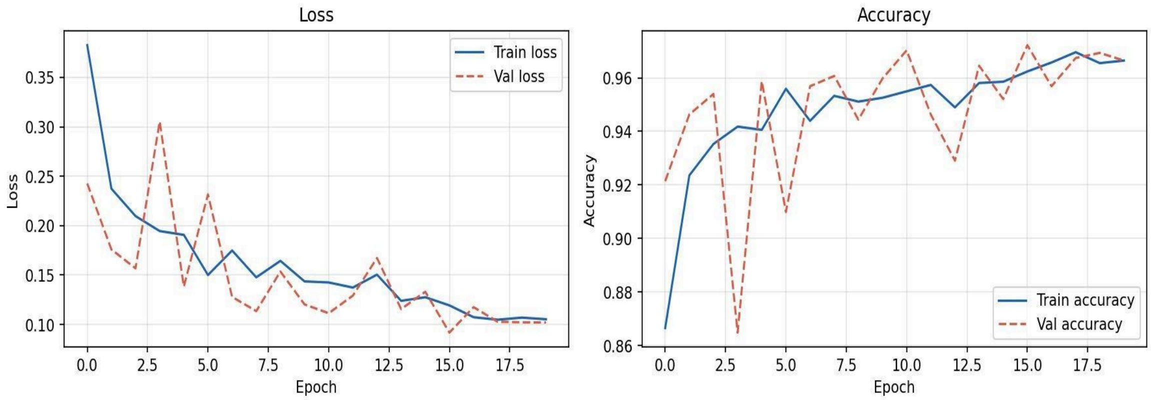
Parameter	Value
Backbone	ResNet50 (ImageNet weights)
Phase 1 Learning Rate	0.001
Phase 2 Learning Rate	0.00001
Epochs per Phase	20
Activation (Output)	Sigmoid
Loss Function	Binary Cross-Entropy
Optimizer	Adam

3. Training Results

3.1 Phase 1 — Feature Extraction

During Phase 1, the training loss decreased steadily from 0.38 to approximately 0.10. Validation accuracy showed some oscillation, which is typical when the backbone is frozen and only the head is being optimized. Both train and validation accuracy converged near 96–97% by the end of Phase 1.

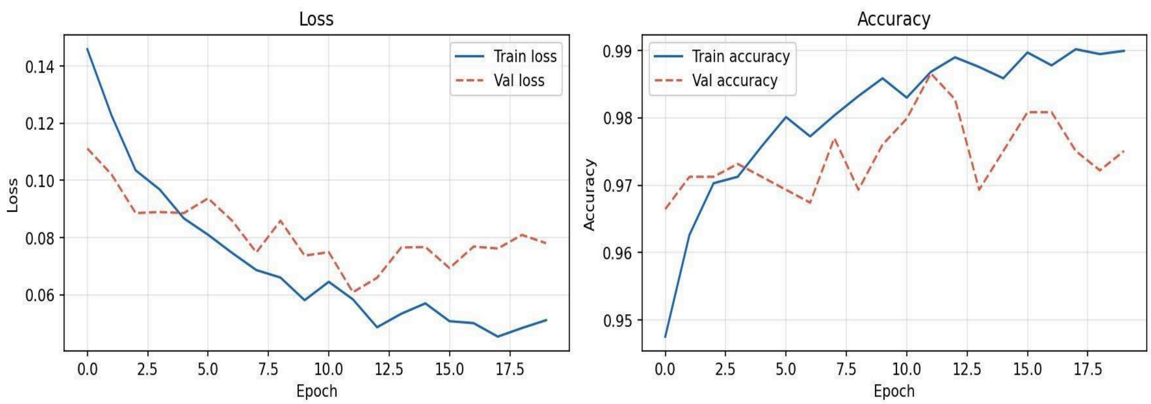
Phase 1 Training Curves (Feature Extraction)



3.2 Phase 2 — Fine-tuning

Phase 2 fine-tuning further reduced training loss to approximately 0.05. Training accuracy reached ~99%, with validation accuracy stabilizing around 97–98%. The gap between training and validation performance is modest, indicating good generalization without severe overfitting.

Phase 2 Training Curves (Fine-tuning)



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4. Test Set Evaluation

4.1 Overall Performance Metrics

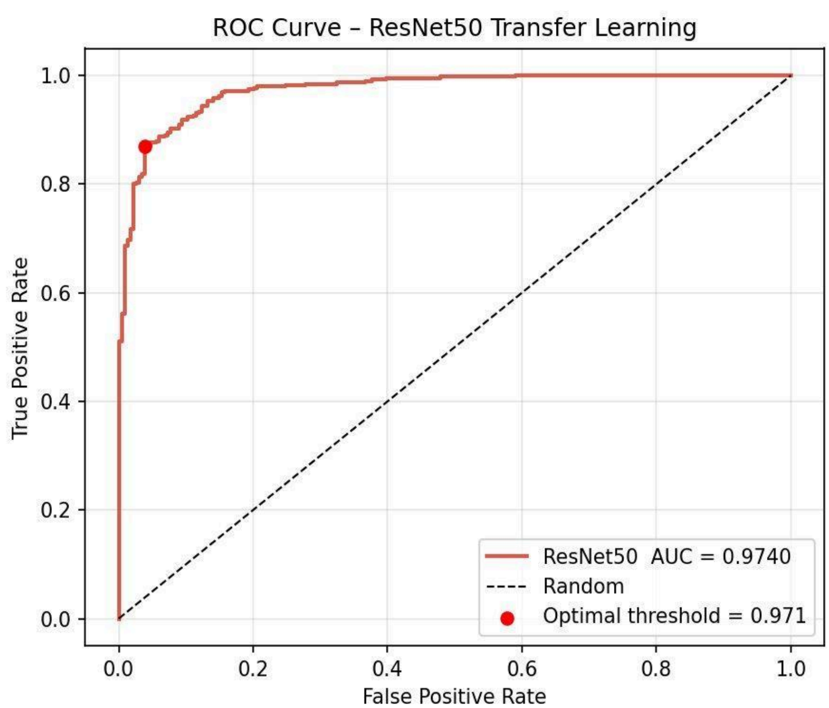
The model was evaluated on a held-out test set. Two decision thresholds were analyzed: the default threshold of 0.5 and an optimized threshold of 0.971 determined via ROC curve analysis.

Metric	Default (t=0.5)	Optimal (t=0.971)	Difference
Accuracy	92.15%	90.38%	-1.77%
Precision	92.31%	97.41%	+5.10%

Metric	Default (t=0.5)	Optimal (t=0.971)	Difference
Pneumonia Recall	95.38%	86.92%	-8.46%
Normal Recall	86.75%	96.15%	+9.40%
F1 Score	93.82%	91.87%	-1.95%
ROC-AUC	97.40%	97.40%	—

4.2 ROC Curve Analysis

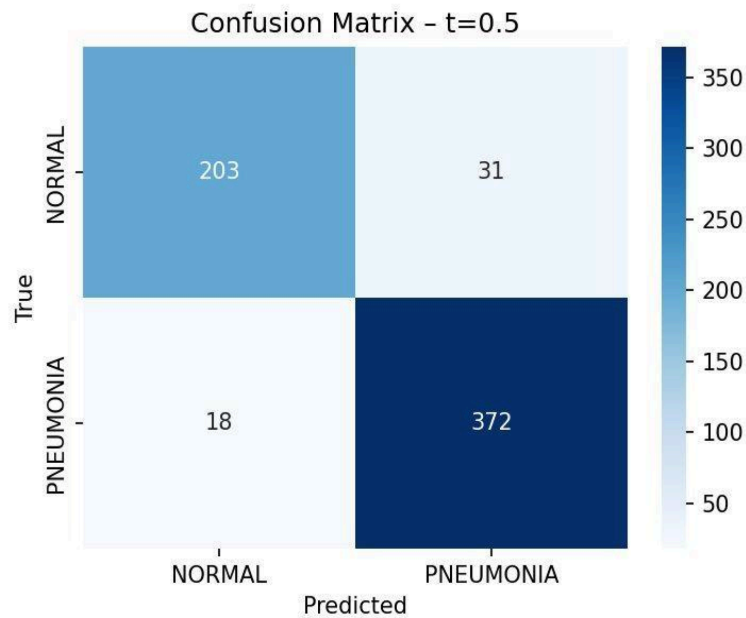
The model achieved an ROC-AUC of 0.974, demonstrating excellent discriminative ability between normal and pneumonia cases. The optimal threshold of 0.971 was identified as the point on the ROC curve closest to the top-left corner (maximizing true positive rate while minimizing false positive rate).



5. Confusion Matrix Analysis

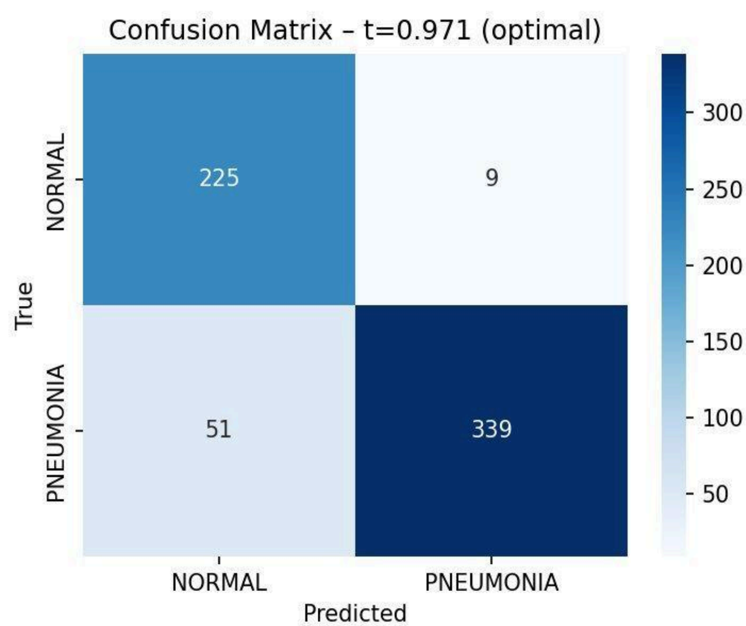
5.1 Default Threshold (t = 0.5)

At the default threshold, the model correctly classified 203 NORMAL and 372 PNEUMONIA cases. It produced 31 false positives (NORMAL predicted as PNEUMONIA) and 18 false negatives (PNEUMONIA predicted as NORMAL). The false negative count is critical in a medical context, as missed pneumonia cases carry serious clinical risk.



5.2 Optimal Threshold (t = 0.971)

At the optimal threshold of 0.971, the model significantly reduced false positives from 31 to 9, improving Normal Recall from 86.75% to 96.15%. This comes at the cost of increased false negatives (18 to 51). This trade-off prioritizes specificity over sensitivity suitable for screening scenarios where unnecessary treatment must be minimized.



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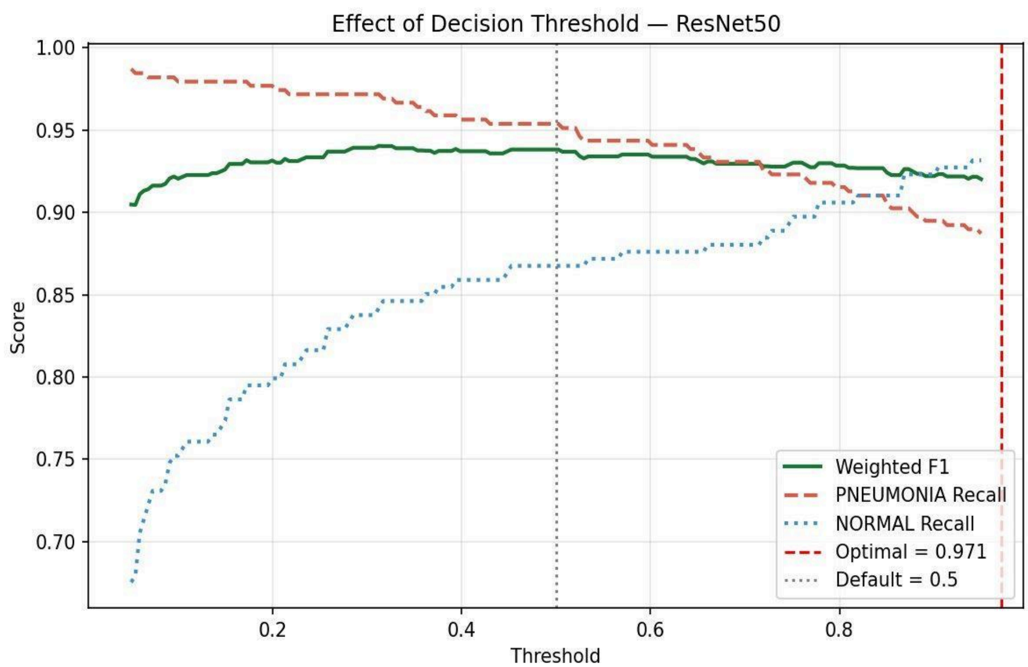
6. Decision Threshold Analysis

The threshold analysis plot illustrates how varying the decision threshold affects Weighted F1, Pneumonia Recall, and Normal Recall simultaneously. The default threshold (0.5) and optimal threshold (0.971) are both marked.

Key observations:

- Pneumonia recall decreases as the threshold increases, since the model becomes more conservative in classifying positive cases.
- Normal recall increases as the threshold rises, correctly rejecting more true-negative cases.
- Weighted F1 remains relatively stable between thresholds 0.3 and 0.8, peaking near 0.5, suggesting the default threshold is well-balanced for general use.
- The optimal threshold at 0.971 is appropriate when the clinical priority is to avoid over-diagnosis.



7. Model Comparison

The ResNet50 transfer learning model was compared against the custom CNN (Variant B) baseline. Transfer learning outperforms the custom CNN across all four evaluated metrics:

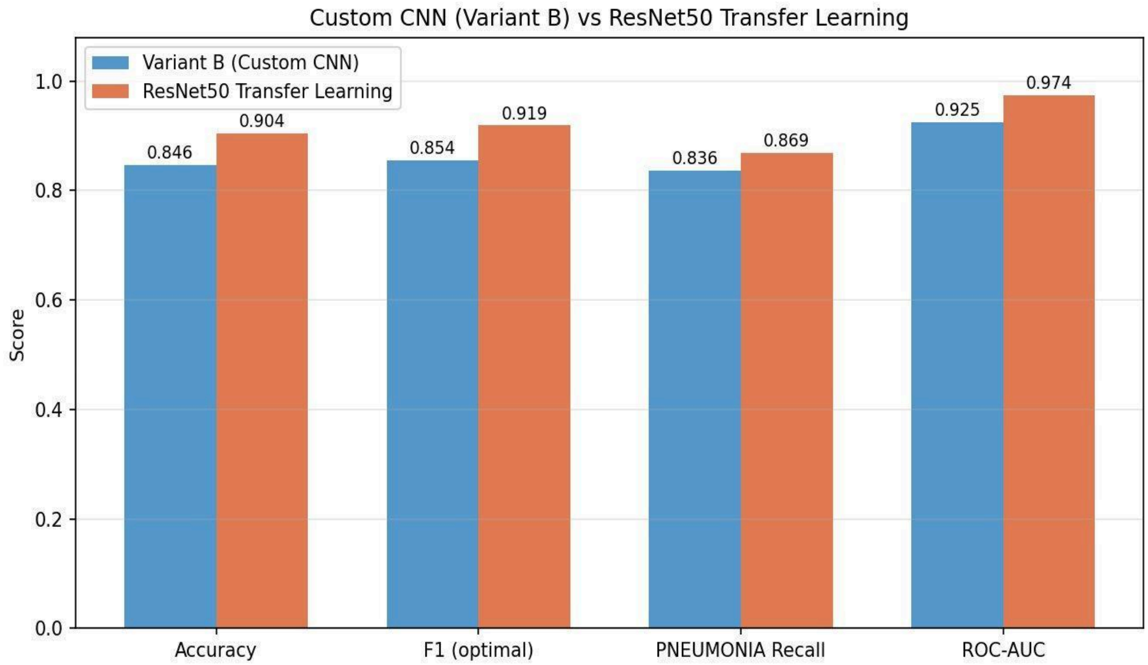


Figure 7: Custom CNN (Variant B) vs. ResNet50 Transfer Learning — Performance Comparison

Metric	Variant B (CNN)	ResNet50 (TL)	Improvement
Accuracy	84.6%	90.4%	+5.8%
F1 Score	85.4%	91.9%	+6.5%
Pneumonia Recall	83.6%	86.9%	+3.3%
ROC-AUC	92.5%	97.4%	+4.9%

8. Discussion

8.1 Threshold Selection

The choice between the default and optimal threshold depends on the clinical use case:

- Default threshold (t=0.5): Higher overall accuracy (92.15%) and F1 (93.82%). Pneumonia recall of 95.38% ensures most pneumonia cases are caught. Recommended for primary screening where sensitivity is paramount.
- Optimal threshold (t=0.971): Significantly higher precision (97.41%) and Normal Recall (96.15%). Produces far fewer false alarms (9 vs. 31). Recommended for confirmatory diagnosis or in settings where false positives have high cost (e.g., unnecessary hospitalisations).

8.2 Transfer Learning Advantages

The significant improvement of ResNet50 over the custom CNN demonstrates the power of transfer learning in medical imaging. ImageNet pre-training provides rich low-level feature detectors (edges, textures) that generalize well to X-ray images, even though the domains differ substantially. With limited labeled medical data, this approach is particularly valuable.

8.3 Limitations

- Phase 2 training shows a growing gap between train (~99%) and validation (~97%) accuracy, suggesting mild overfitting that could be addressed with additional regularization or data augmentation.
- The dataset class imbalance (more PNEUMONIA than NORMAL samples) may bias the model; class weighting or oversampling should be considered for future iterations.
- The high optimal threshold (0.971) is unusual and may indicate the model outputs poorly calibrated probabilities. Platt scaling or isotonic regression could improve calibration.

9. Conclusion

The ResNet50 transfer learning model achieves strong performance on the pneumonia detection task, with an ROC-AUC of 0.974 and a default-threshold accuracy of 92.15%. The two-phase training strategy effectively leveraged pre-trained features while adapting them to the medical imaging domain.

The model outperforms the custom CNN baseline by 5.8 percentage points in accuracy and 4.9 percentage points in ROC-AUC, confirming that transfer learning is the preferred approach for this classification problem given the available data.

For clinical deployment, the default threshold (t=0.5) is recommended for initial screening, with the possibility of applying the optimal threshold (t=0.971) as a secondary confirmatory filter to reduce unnecessary referrals.

Key Results Summary

ROC-AUC: 0.974 | Accuracy (t=0.5): 92.15% | F1 Score: 93.82%

Pneumonia Recall: 95.38% | Optimal Threshold: 0.971 | Test Loss: 0.2833